

Practice Final Test-Part I (Version 2)

Name
Date
Course

- I) This exam covers the first half of the course (everything up to test II).
- II) Write your name, the date, and the course title (i.e. Precalculus).
- III) Show all work.
- IV) Draw a rectangle around your final answer for each problem.
- V) You may use a calculator for counting problems only, not for graphing.

① The probability that a randomly selected person will vote for candidate "A" is $\frac{1}{2}$. The probability that a randomly selected person will vote for candidate "B" is $\frac{2}{7}$. What is the probability that a randomly selected person will vote for either candidate "A" or candidate "B"?

② A polynomial function is written below in factored form. Write the x-intercepts of the corresponding graph and determine if the function crosses or touches the x-axis at each intercept.

$$f(x) = -3(x+1)^7(x-\sqrt{10})^6(x+8.7)^2(x-8)^{11}$$

③ i) Indicate the end behavior of the function below.

$$f(x) = -x^3 - 6x^2 - 4x + 17$$

ii) State the maximum number of turning points of the graph of the function.

④ Solve. Write the solutions graphically and in interval notation.

$$0 \geq 2x^3 + x^2 - 21x$$

⑤ You have 7 different pairs of shoes, 6 different t-shirts, 4 different pairs of pants, and 3 different hats. How many outfit variations are possible?

⑥ 10 scientists must be chosen to study the coral reef in Australia. If there are 15 scientists willing to go, how many possible outcome variations are there?

⑦ Find the value of the expression below.

$$\sum_{n=1}^{\infty} (-0.77)^{n-1}$$

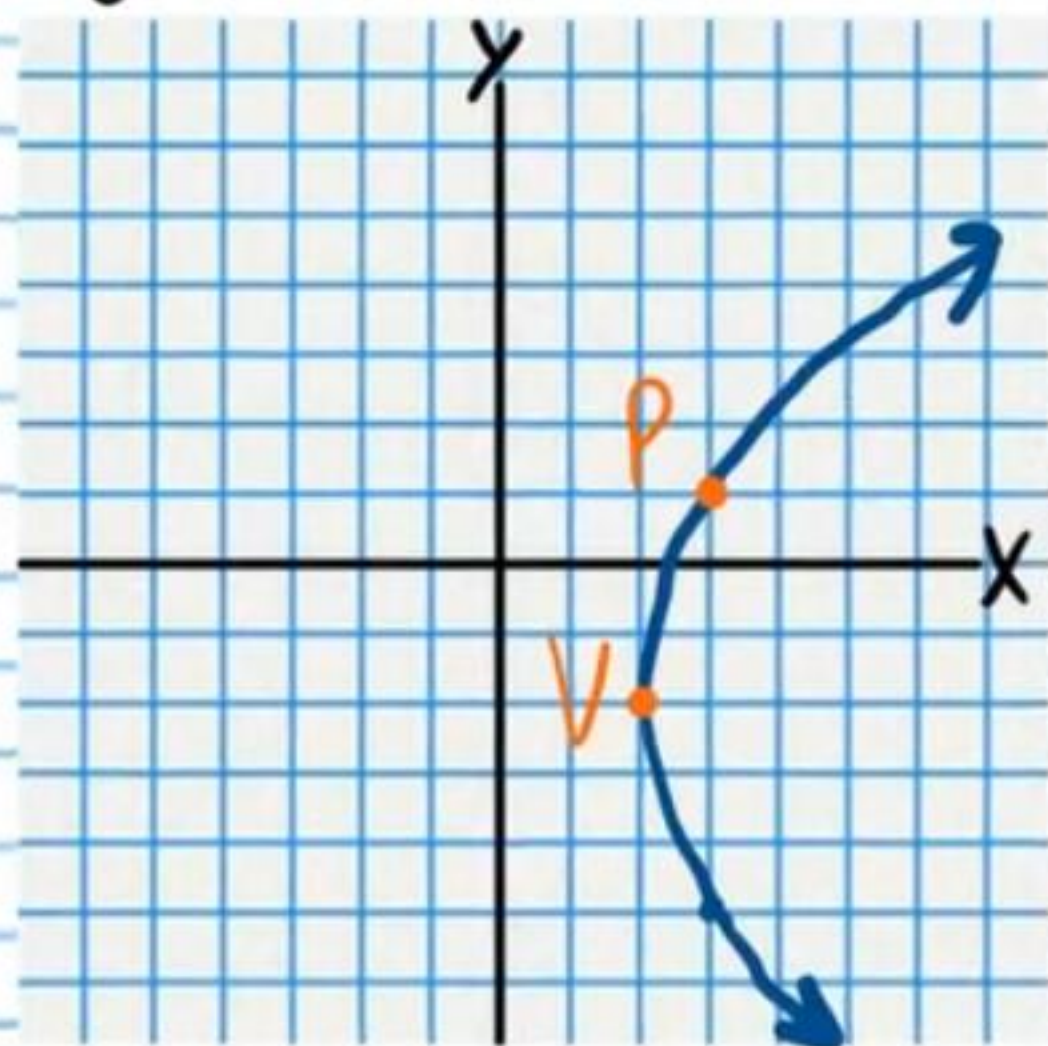
⑧ i) Find the 41st term of the sequence below.

$$-18, -8, 2, 12, \dots$$

ii) Find the sum of the first 41 terms of the sequence.

⑨ Find an expression for the n^{th} term of the sequence in problem #8. Write in terms of " n ," not in recursive form.

⑩ "P" is a point on the parabola and "V" is the vertex. Write the equation of the graph below.



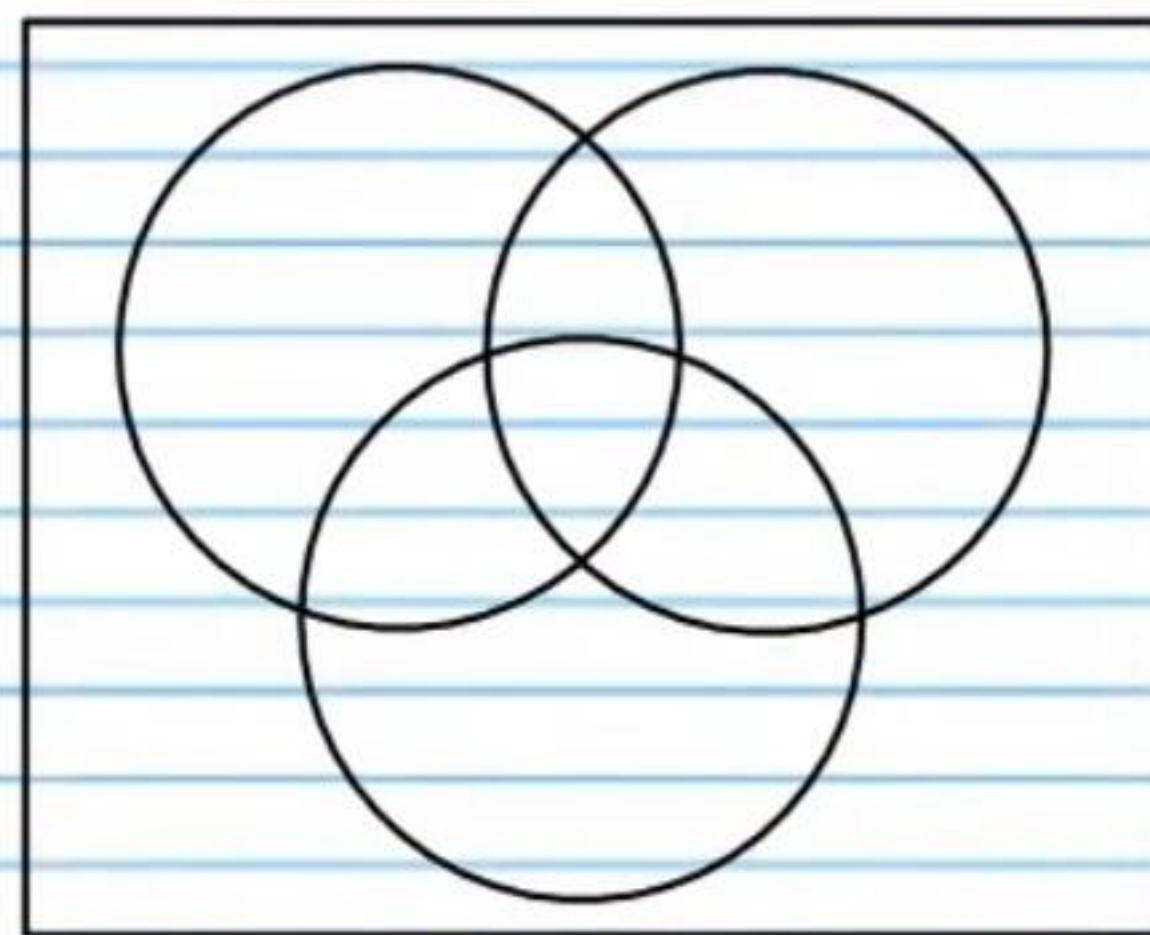
⑪ Use the Binomial Theorem to expand the following expression.
 $(2x + 3y)^5$

⑫ Find the value of the expression below.

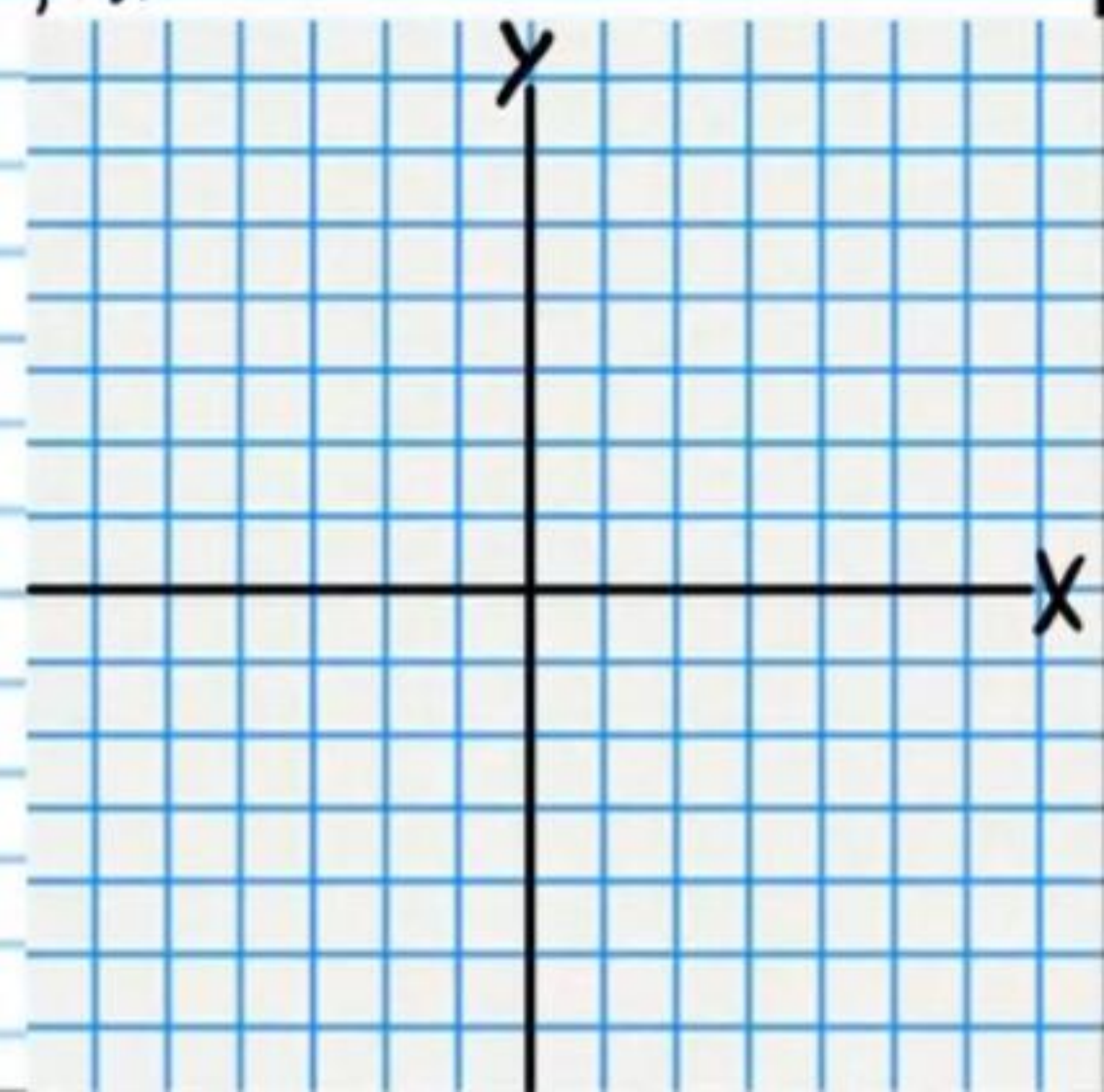
$$\sum_{n=0}^{32} 3\left(\frac{2}{3}\right)^n$$

⑬ There are 7 green marbles and 6 red marbles in a bag. 3 marbles are chosen at random. What is the probability that at least one marble will be red?

⑭ There are 116 farms in the county of Petersburg. 52 of them have cows, 46 of them have chickens, and 29 of them have horses and chickens. 20 of them have horses and cows. 16 of them have cows and chickens. 9 of them have all three. 19 farms do not have cows, chickens, or horses. How many farms have horses only?

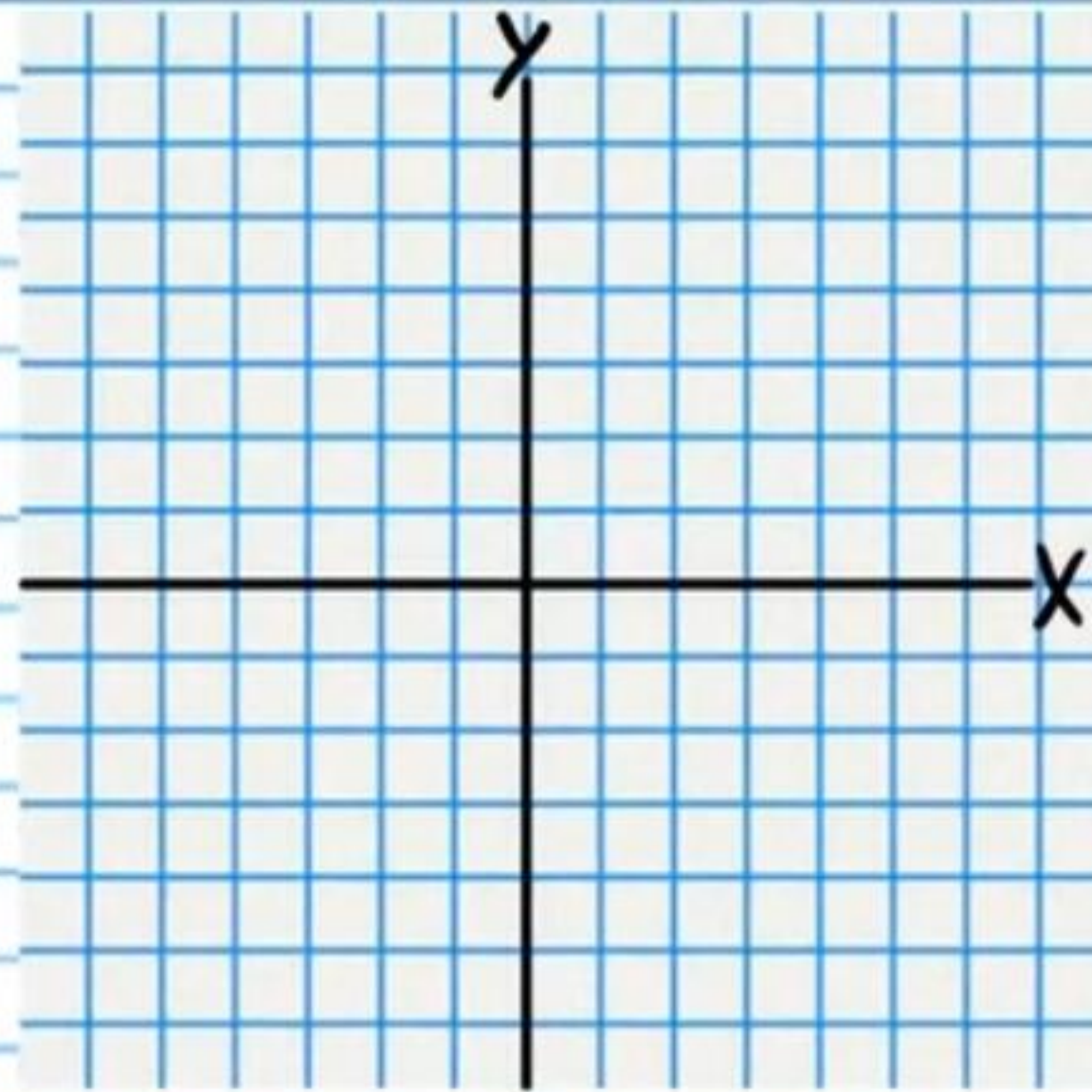


⑮ Find an equation of the ellipse with the following characteristics. *Graph the equation.* Center $(2, 3)$, Vertex $(-3, 3)$, containing the point $(2, 0)$.



⑩ Graph the equation below.

$$x^2 + y^2 + 8x + 8y + 23 = 0$$



⑪ Write a polynomial function in standard form with real coefficients that has the following zeros, leading coefficient (LC), and degree (D).

$$\text{Zeros: } 3 + 2i, -2, 1 \quad \text{LC} = 2 \quad \text{D} = 4$$

⑫ Solve. Write the solutions graphically and in interval notation.

$$5x^2 + 2x > 2$$

①9 Find the roots of the polynomial function (both real and imaginary).

$$f(x) = 9x^4 + 20x^2 - 125$$

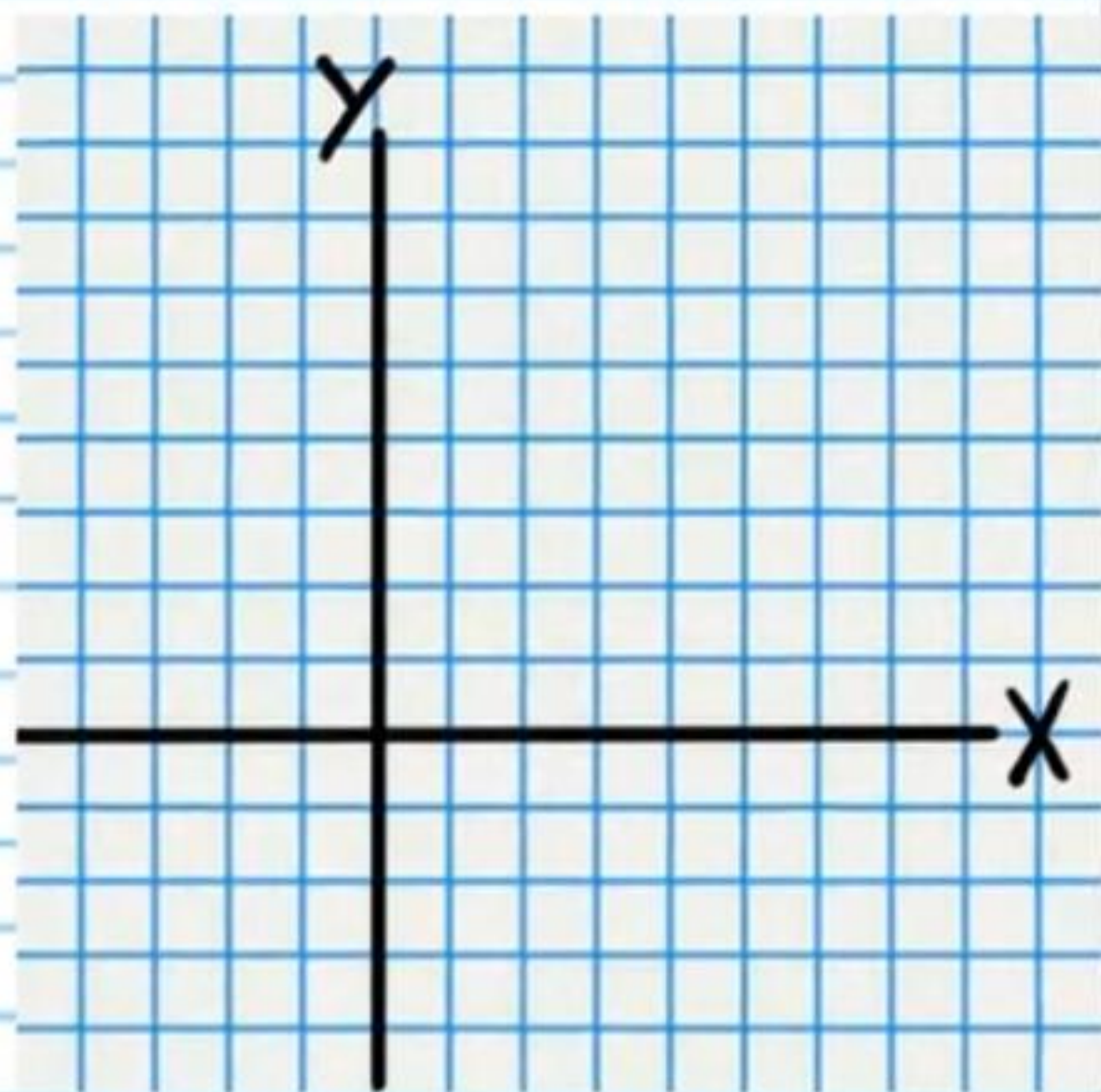
②0 i) Divide.

$$(4x^2 - 2x^4 + 4) \div (4x - x^2)$$

ii) Can you use synthetic division? Explain.

21) Graph the equation below.

$$y^2 - 5x^2 - 6y + 30x - 41 = 0$$



22) Solve. Write the solutions graphically and in interval notation.

$$2 \leq \frac{x+1}{x-2}$$

②③ Use the Rational Zeros Theorem to Solve the equation below.

$$9x^5 - 49x^3 + 4x^2 + 52x - 16 = 0$$

②④ Graph the following function. Draw asymptotes as dotted lines and write their equations.

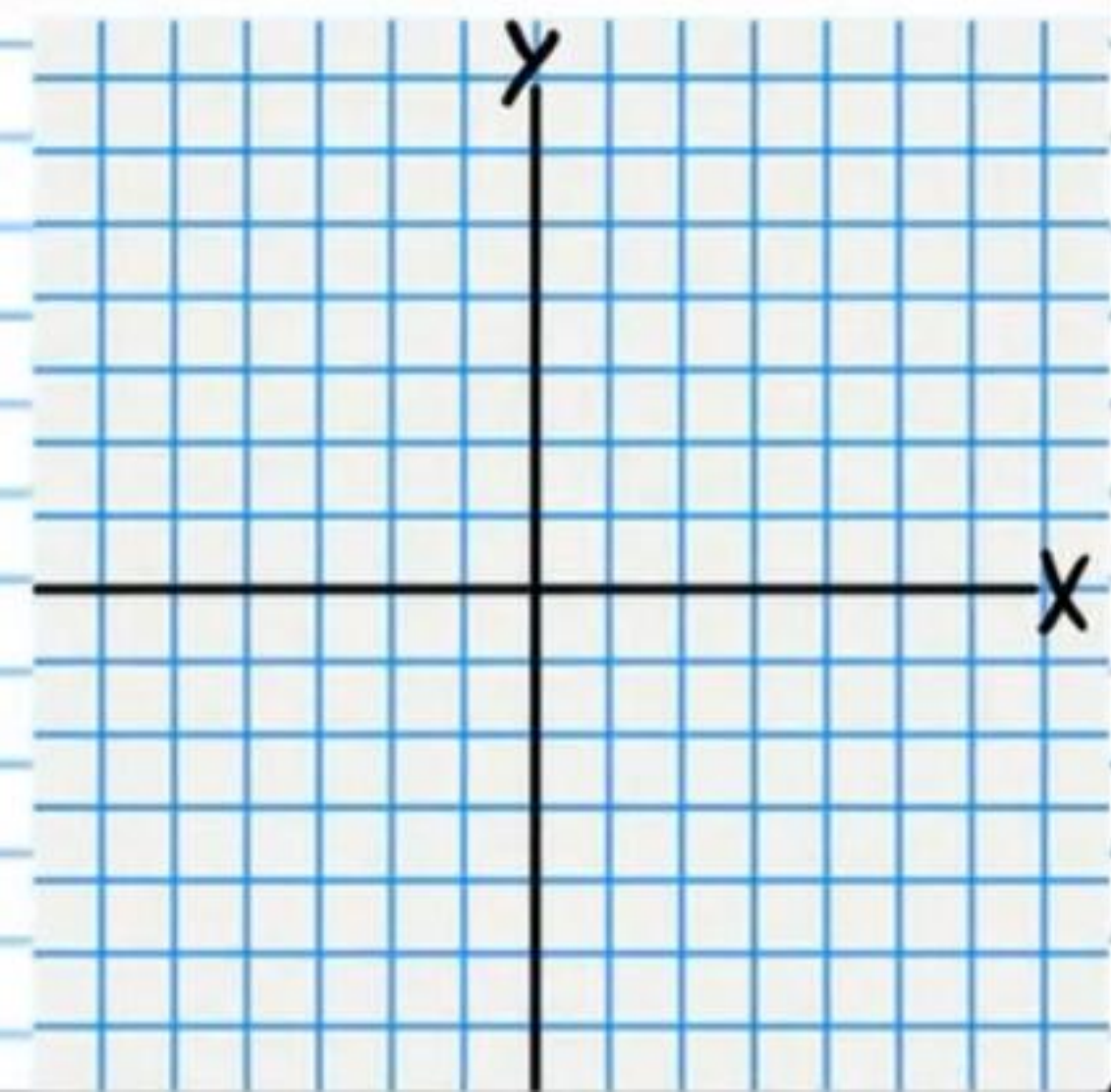
$$f(x) = \frac{x^3 + 64}{x^3}$$

Horizontal Asymptote Equation

Vertical Asymptote Equation(s) Sign Chart

Y-Intercept

X-Intercept



②5 Graph the following function. Draw asymptotes as dotted lines and write their equations.

$$f(x) = -\frac{x^4 - 81}{x^3 + 2x^2 - 9x - 18}$$

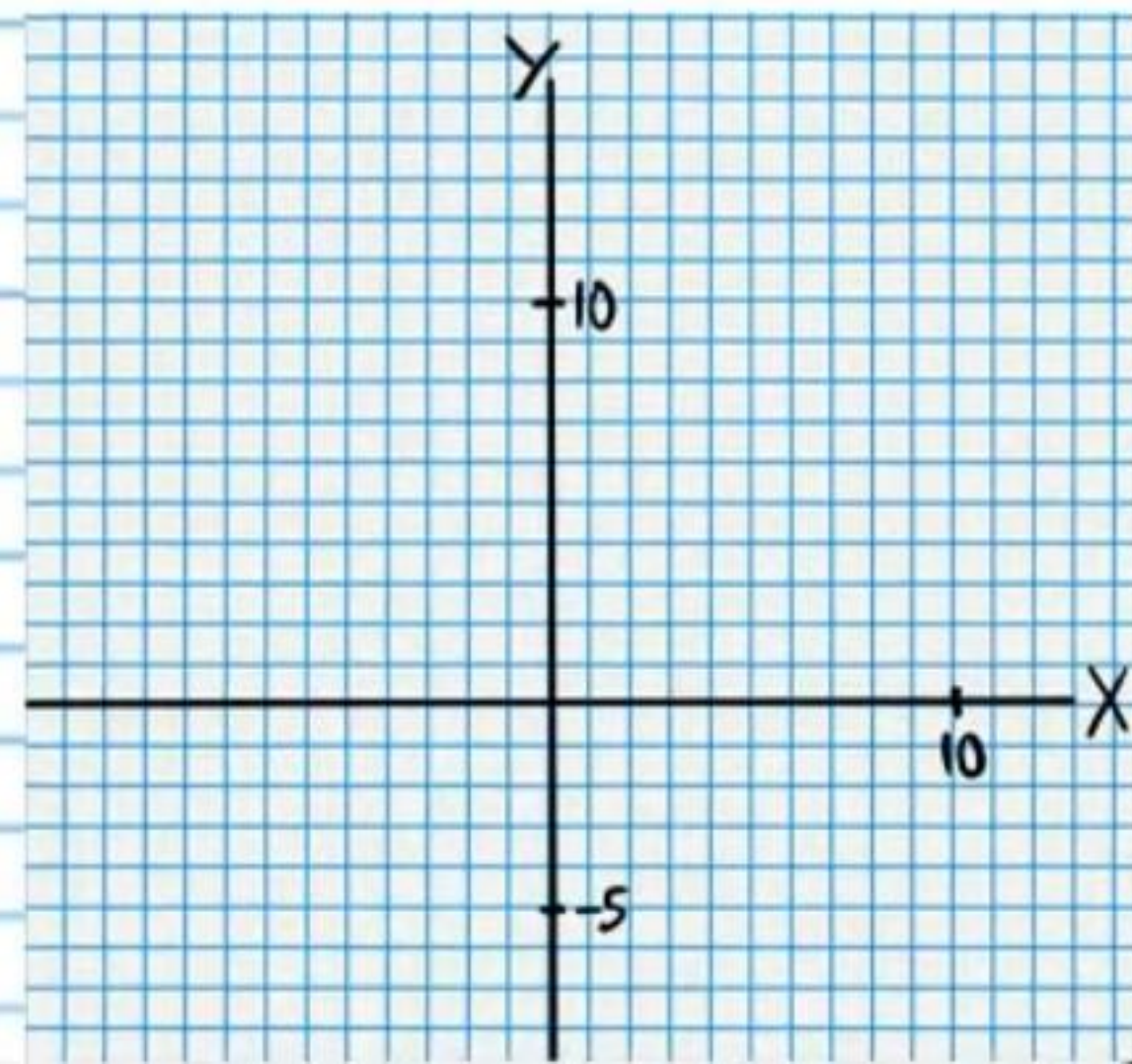
Oblique Asymptote Equation

Hole(s)

Vertical Asymptote Equation(s) Sign Chart

Y-Intercept

X-Intercept



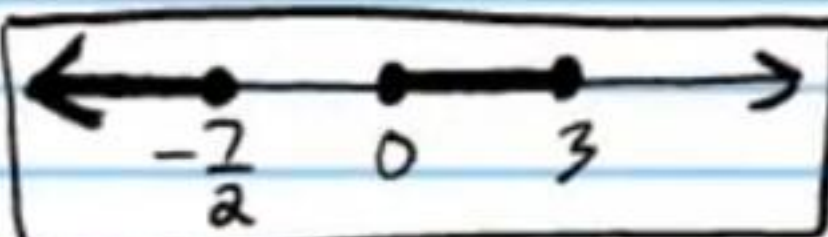
Answers (Version 2)

① $\frac{11}{14}$

② $-1 \Rightarrow$ Crosses
 $\sqrt{10} \Rightarrow$ Touches
 $-8.7 \Rightarrow$ Touches
 $8 \Rightarrow$ Crosses

③ i) As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$
 As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

ii) 2

④ 

$(-\infty, -\frac{7}{2}] \cup [0, 3]$

⑤ 504 outcome variations

⑥ 3,003 outcome variations

⑦ $\frac{100}{177}$ or $0.56497\dots$

⑧ i) $a_{41} = 382$ ii) $S_{41} = 7,462$

⑨ $a_n = 10n - 28$ or $a_n = -28 + 10n$

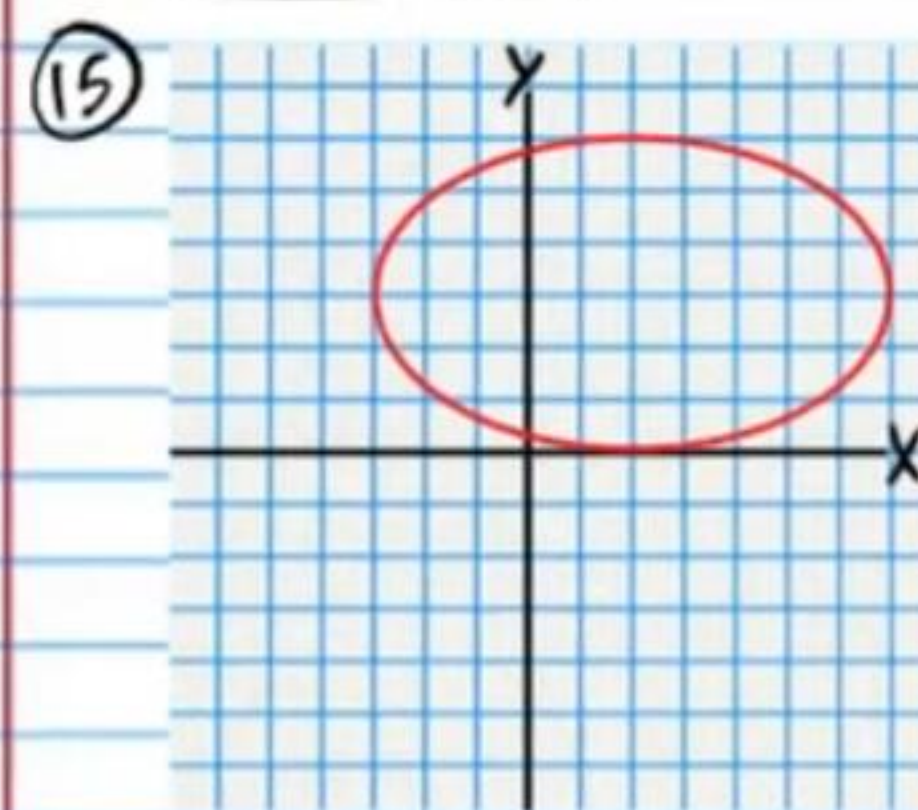
⑩ $\frac{1}{9}(y+2)^2 = x-2$

⑪ $32x^5 + 240x^4y + 720x^3y^2 + 1080x^2y^3 + 810xy^4 + 243y^5$

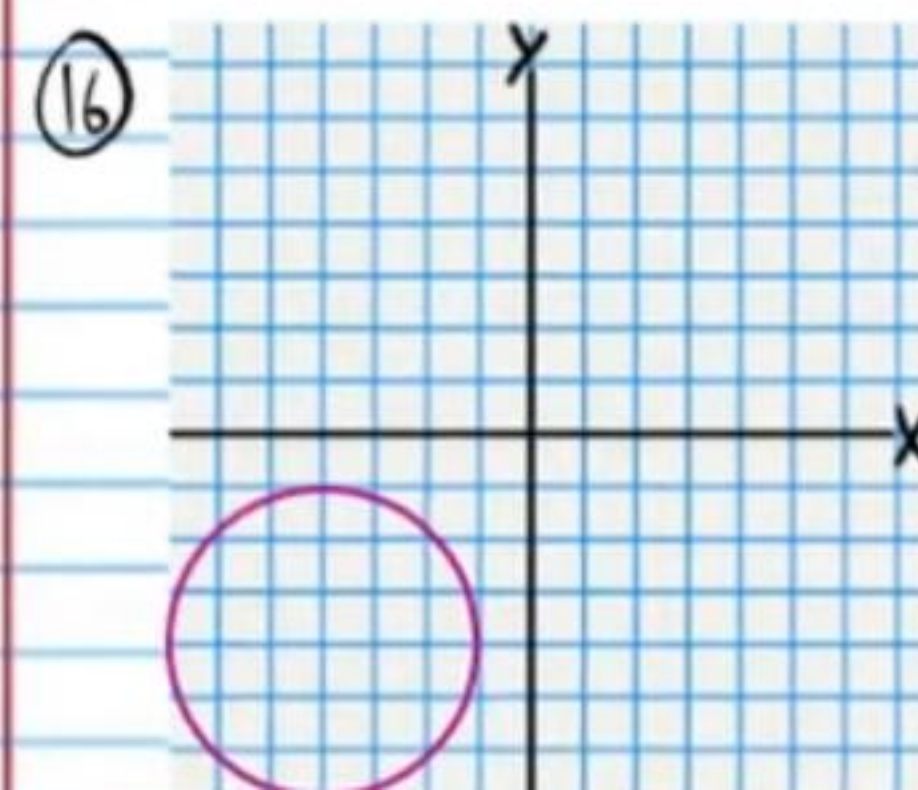
⑫ $9(1 - (\frac{2}{3})^{33})$

⑬ $\frac{251}{286}$

⑭ 15 farms



$\frac{(x-2)^2}{25} + \frac{(y-3)^2}{9} = 1$



(17) $2x^4 - 10x^3 + 10x^2 + 50x - 52$

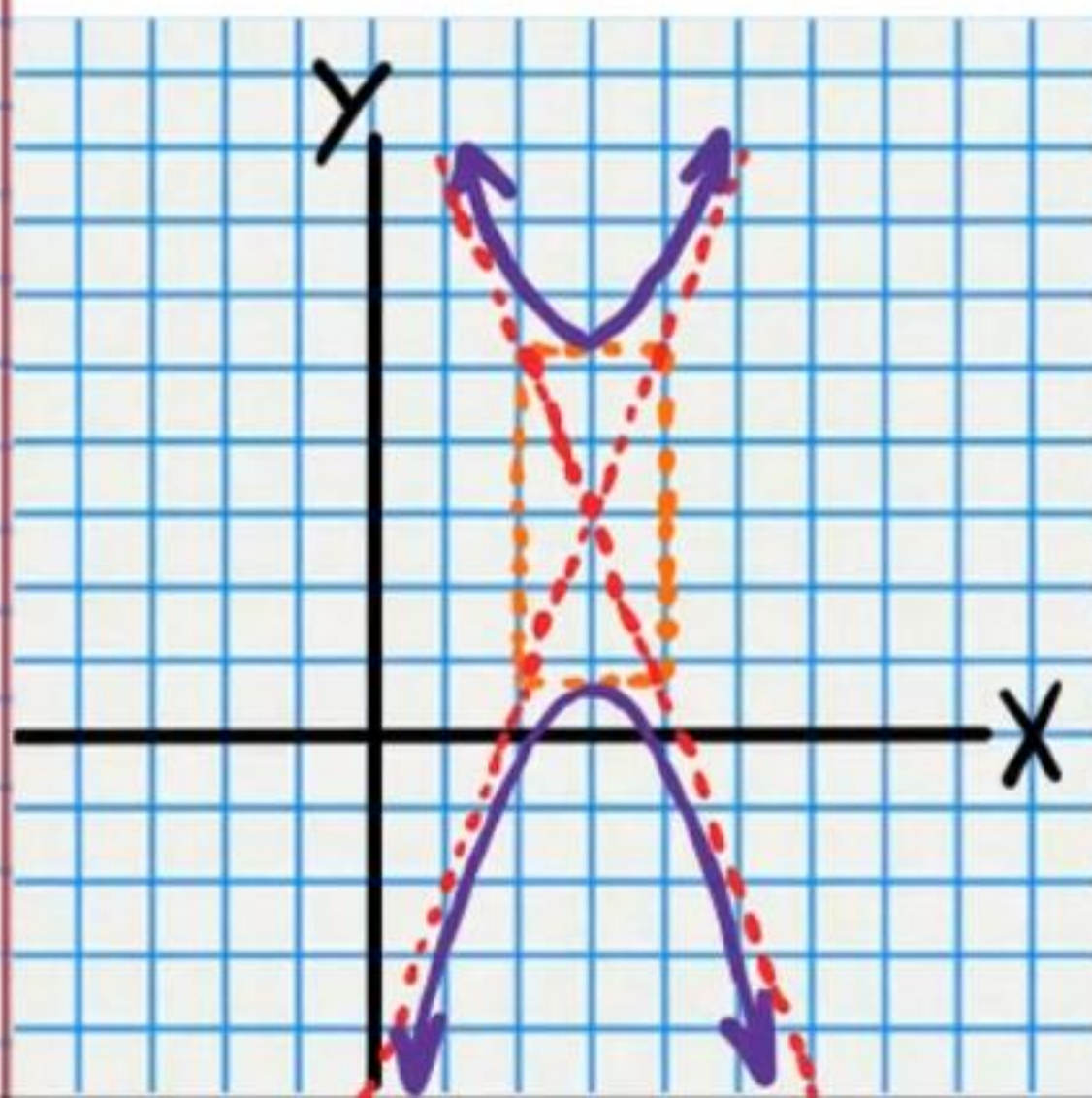
(18) $\left(-\infty, \frac{-1-\sqrt{11}}{5} \right) \cup \left(\frac{-1+\sqrt{11}}{5}, \infty \right)$

(19) $\frac{5}{3}, -\frac{5}{3}, i\sqrt{5}, -i\sqrt{5}$

(20) i) $2x^2 + 8x + 28 + \frac{-112x + 4}{-x^2 + 4x}$

ii) No. The divisor is not of the form $x-a$.

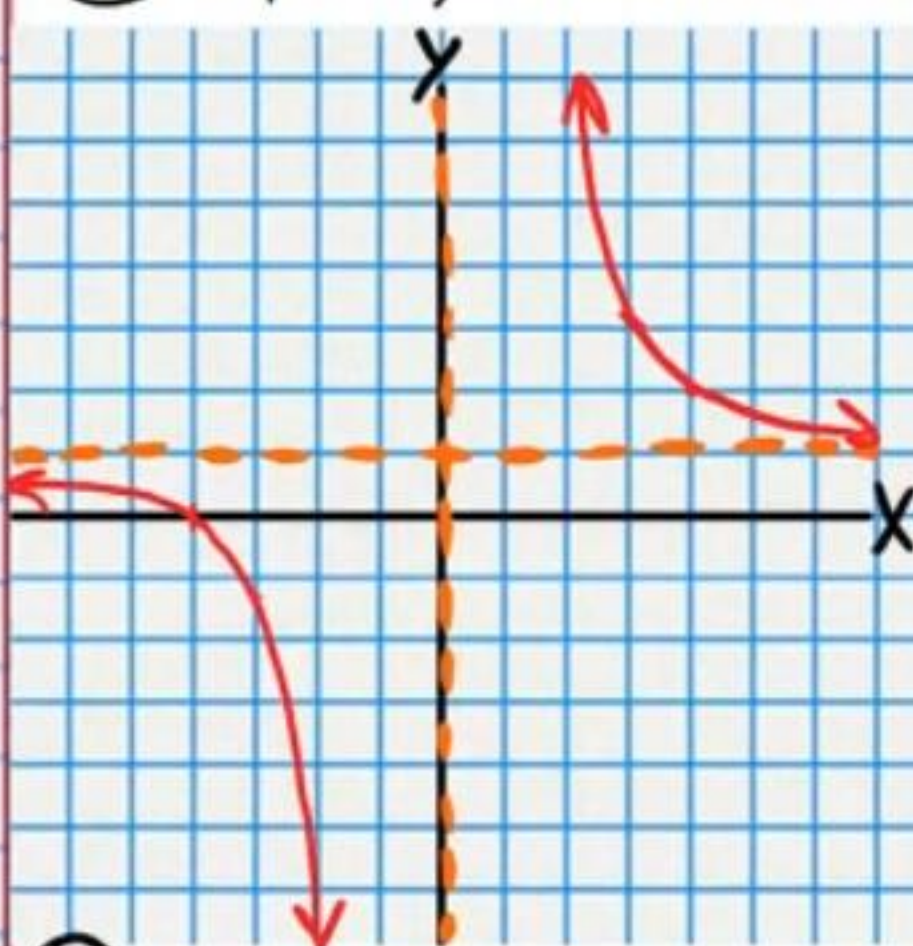
(21)



(22) $[2, 5]$

(23) $x = 1, 2, -2, -\frac{4}{3}, \frac{1}{3}$

(24) $y=1, x=0$



(25) $x = -2, y = -x + 2$

